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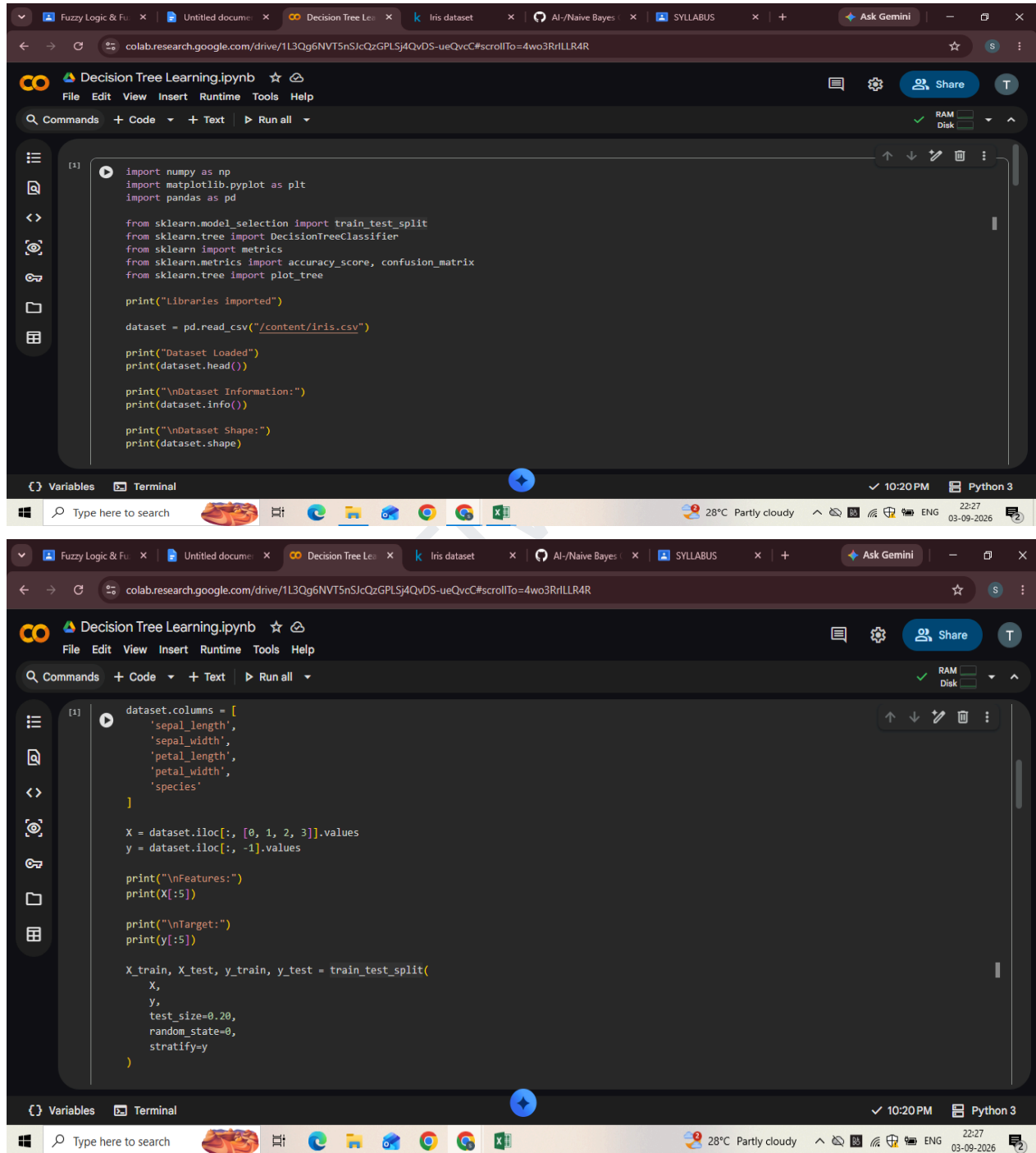
AIM: Decision Tree Learning

Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.

Evaluate the accuracy and effectiveness of the decision tree on test data.

Visualize and interpret the generated decision tree.

INPUT:



```
[1] import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn import metrics
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.tree import plot_tree

print("Libraries imported")

dataset = pd.read_csv("/content/iris.csv")

print("Dataset Loaded")
print(dataset.head())

print("\nDataset Information:")
print(dataset.info())

print("\nDataset Shape:")
print(dataset.shape)
```

```
[1] dataset.columns = [
    'sepal_length',
    'sepal_width',
    'petal_length',
    'petal_width',
    'species'
]

X = dataset.iloc[:, [0, 1, 2, 3]].values
y = dataset.iloc[:, -1].values

print("\nFeatures:")
print(X[:5])

print("\nTarget:")
print(y[:5])

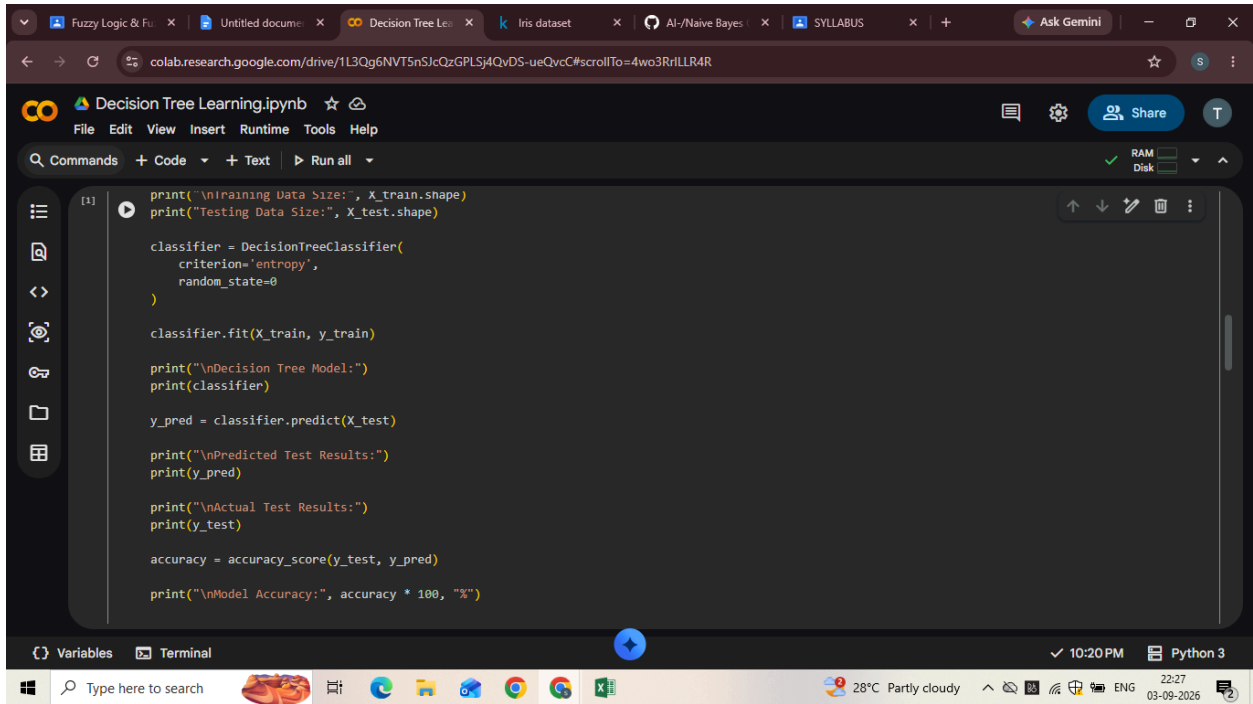
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0,
    stratify=y
)
```

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The screenshot shows a Jupyter Notebook interface with the following code:

```
[1]: print("\nTraining Data Size:", X_train.shape)
     print("\nTesting Data Size:", X_test.shape)

     classifier = DecisionTreeClassifier(
         criterion='entropy',
         random_state=0
     )

     classifier.fit(X_train, y_train)

     print("\nDecision Tree Model:")
     print(classifier)

     y_pred = classifier.predict(X_test)

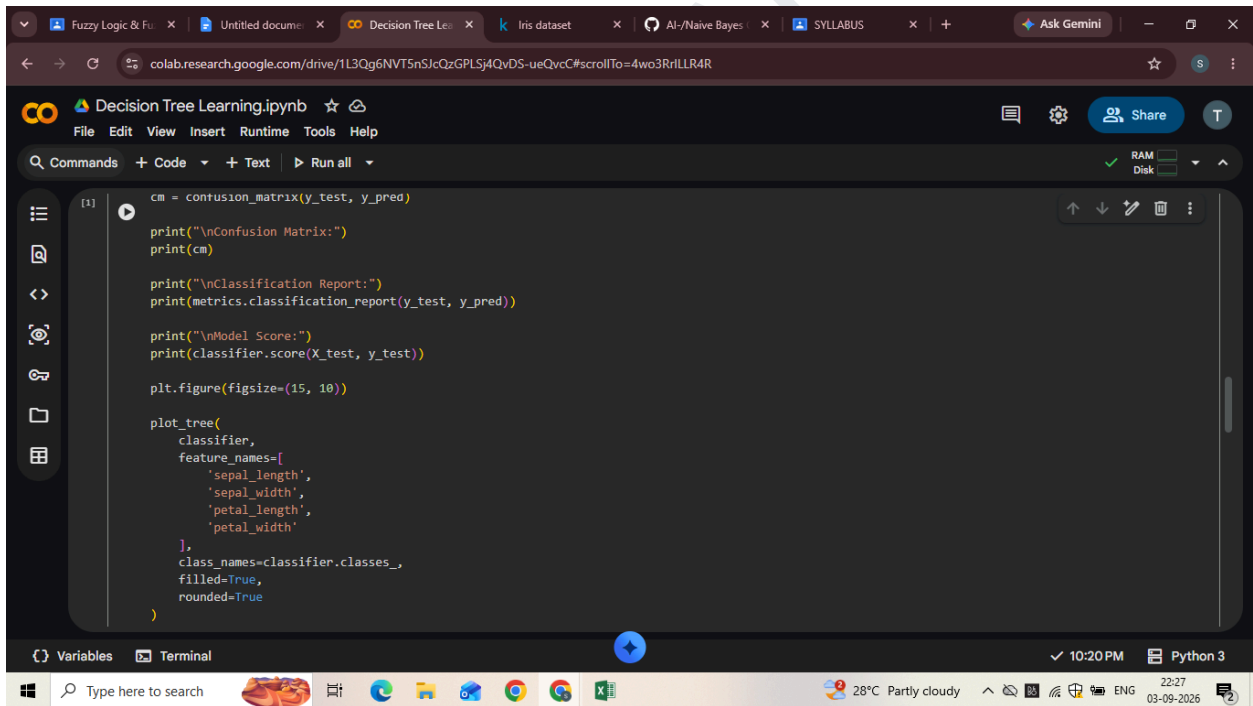
     print("\nPredicted Test Results:")
     print(y_pred)

     print("\nActual Test Results:")
     print(y_test)

     accuracy = accuracy_score(y_test, y_pred)

     print("\nModel Accuracy:", accuracy * 100, "%")
```

The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help), a toolbar with icons for commands, code, text, and running all cells, and a status bar at the bottom showing the time as 10:20 PM and the Python version as Python 3.



The screenshot shows the same Jupyter Notebook with additional code added to the cell:

```
[1]: cm = confusion_matrix(y_test, y_pred)

     print("\nConfusion Matrix:")
     print(cm)

     print("\nClassification Report:")
     print(metrics.classification_report(y_test, y_pred))

     print("\nModel Score:")
     print(classifier.score(X_test, y_test))

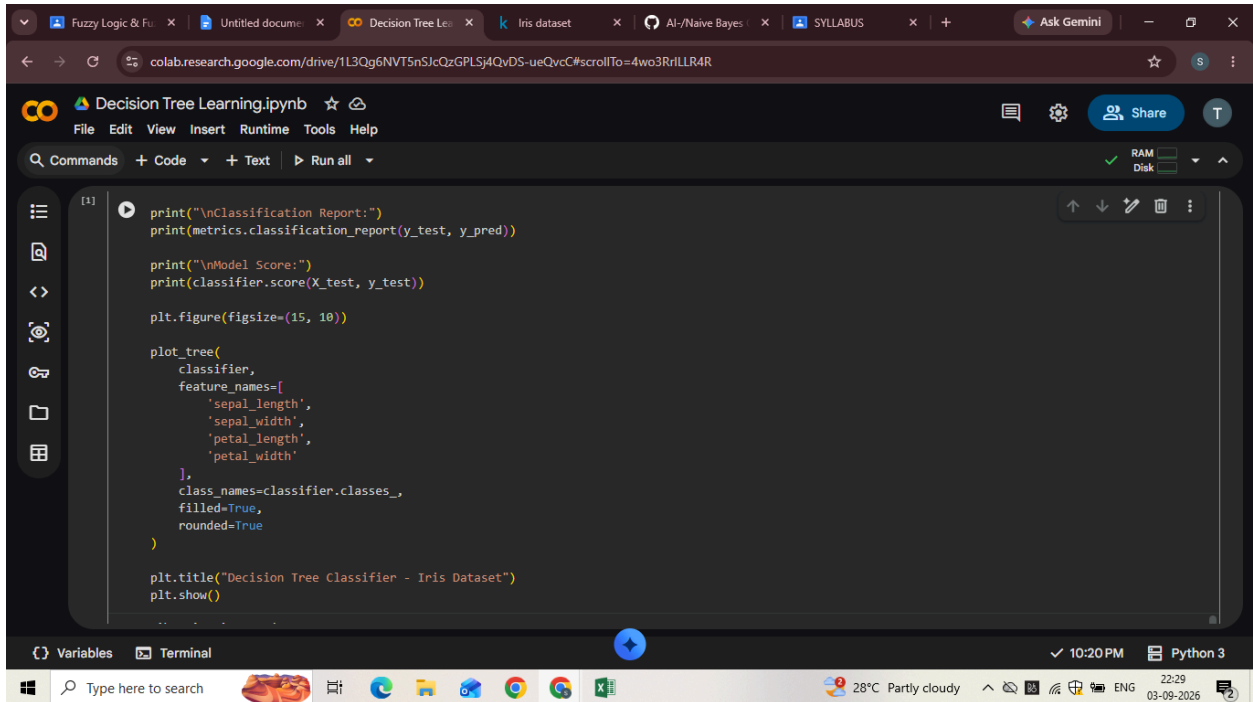
     plt.figure(figsize=(15, 10))

     plot_tree(
         classifier,
         feature_names=[
             'sepal_length',
             'sepal_width',
             'petal_length',
             'petal_width'
         ],
         class_names=classifier.classes_,
         filled=True,
         rounded=True
     )
```

The interface remains the same as the previous screenshot, with the same menu bar, toolbar, and status bar.

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```
[1] print("\nClassification Report:")
print(metrics.classification_report(y_test, y_pred))

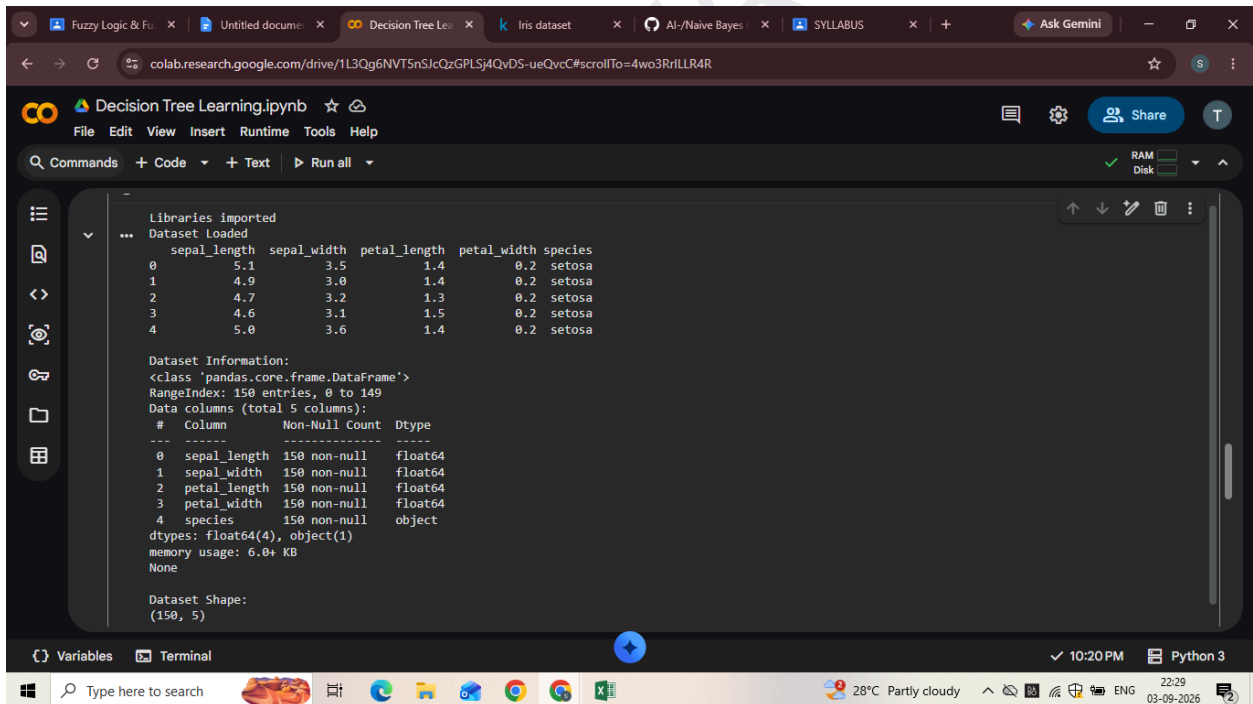
print("\nModel Score:")
print(classifier.score(X_test, y_test))

plt.figure(figsize=(15, 10))

plot_tree(
    classifier,
    feature_names=[
        'sepal_length',
        'sepal_width',
        'petal_length',
        'petal_width'
    ],
    class_names=classifier.classes_,
    filled=True,
    rounded=True
)

plt.title("Decision Tree Classifier - Iris Dataset")
plt.show()
```

OUTPUT:



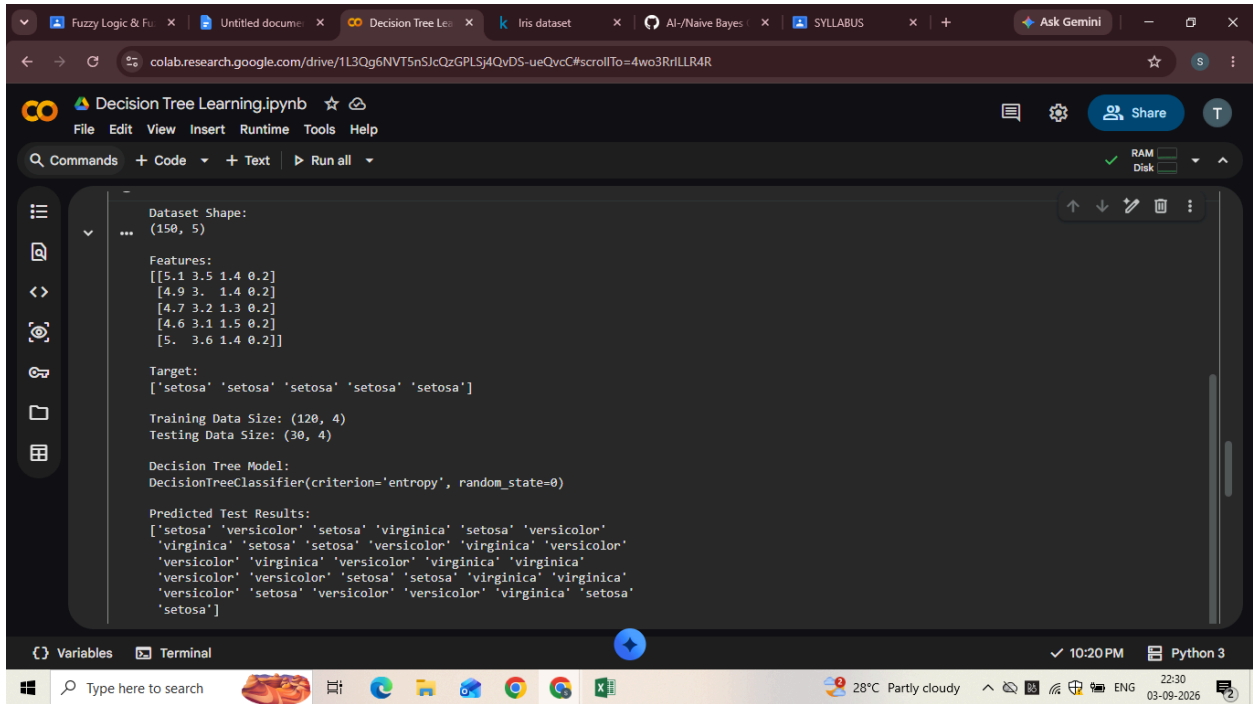
```
Libraries imported
Dataset loaded
  sepal_length  sepal_width  petal_length  petal_width  species
0         5.1         3.5         1.4         0.2  setosa
1         4.9         3.0         1.4         0.2  setosa
2         4.7         3.2         1.3         0.2  setosa
3         4.6         3.1         1.5         0.2  setosa
4         5.0         3.6         1.4         0.2  setosa

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null    float64
1   sepal_width     150 non-null    float64
2   petal_length    150 non-null    float64
3   petal_width     150 non-null    float64
4   species         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None

Dataset Shape:
(150, 5)
```

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The screenshot shows a Jupyter Notebook interface with the following content:

```
Dataset Shape:
(150, 5)

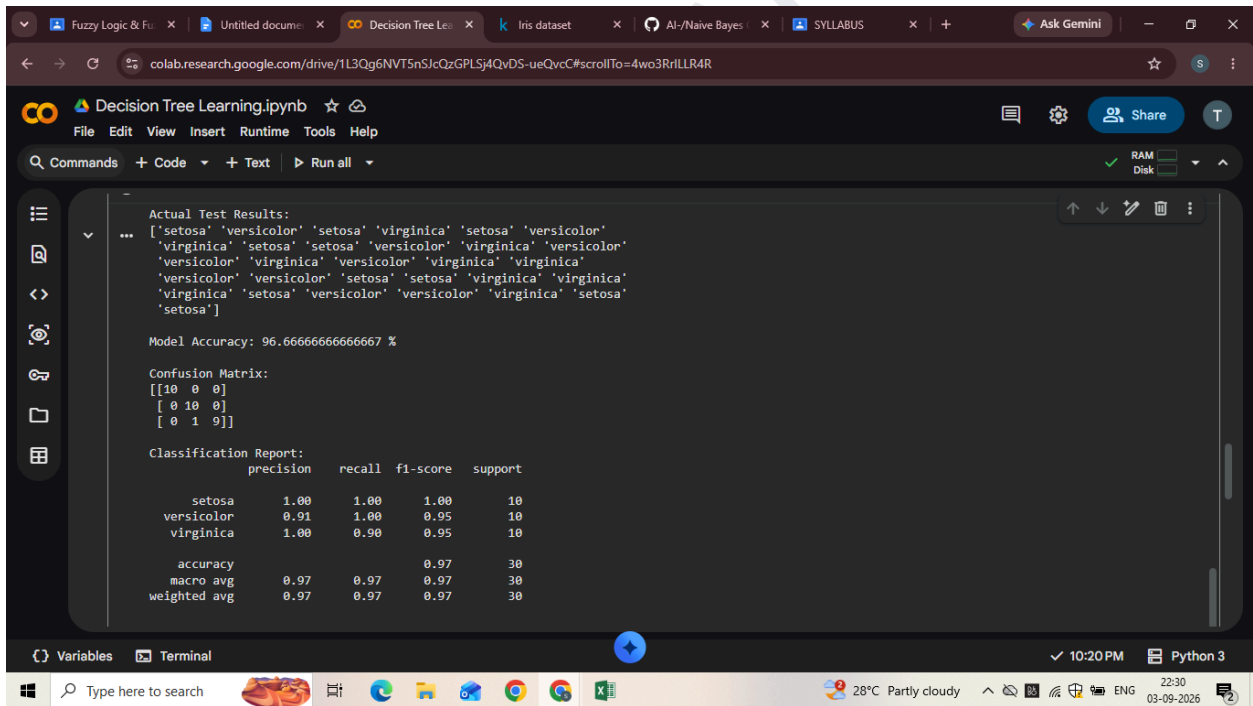
Features:
[[5.1 3.5 1.4 0.2]
 [4.9 3. 1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5. 3.6 1.4 0.2]]

Target:
['setosa' 'setosa' 'setosa' 'setosa' 'setosa']

Training Data Size: (120, 4)
Testing Data Size: (30, 4)

Decision Tree Model:
DecisionTreeClassifier(criterion='entropy', random_state=0)

Predicted Test Results:
['setosa' 'versicolor' 'setosa' 'virginica' 'setosa' 'versicolor'
 'virginica' 'setosa' 'setosa' 'versicolor' 'virginica' 'versicolor'
 'versicolor' 'virginica' 'versicolor' 'virginica' 'virginica'
 'versicolor' 'versicolor' 'setosa' 'setosa' 'virginica' 'virginica'
 'versicolor' 'setosa' 'versicolor' 'versicolor' 'virginica' 'setosa'
 'setosa']
```



The screenshot shows the same Jupyter Notebook interface with the following content:

```
Actual Test Results:
['setosa' 'versicolor' 'setosa' 'virginica' 'setosa' 'versicolor'
 'virginica' 'setosa' 'setosa' 'versicolor' 'virginica' 'versicolor'
 'versicolor' 'virginica' 'versicolor' 'virginica' 'virginica'
 'versicolor' 'versicolor' 'setosa' 'setosa' 'virginica' 'virginica'
 'virginica' 'setosa' 'versicolor' 'versicolor' 'virginica' 'setosa'
 'setosa']

Model Accuracy: 96.66666666666667 %

Confusion Matrix:
[[10  0  0]
 [ 0 10  0]
 [ 0  1  9]]

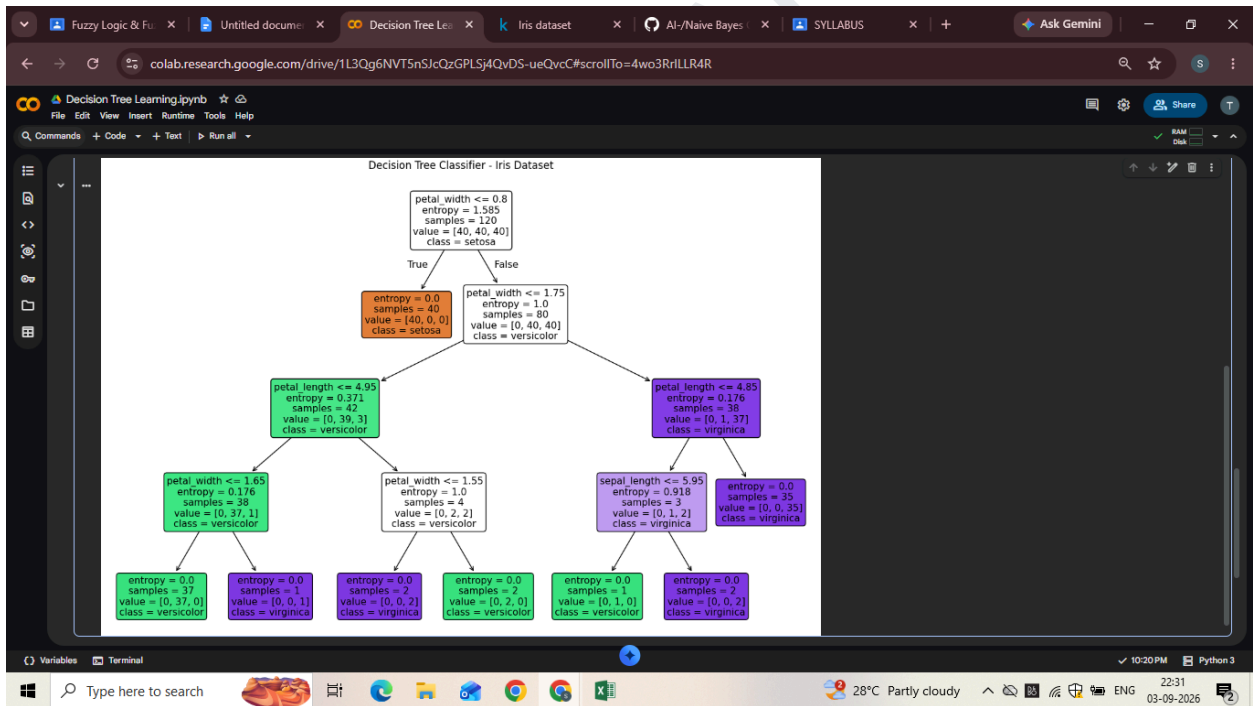
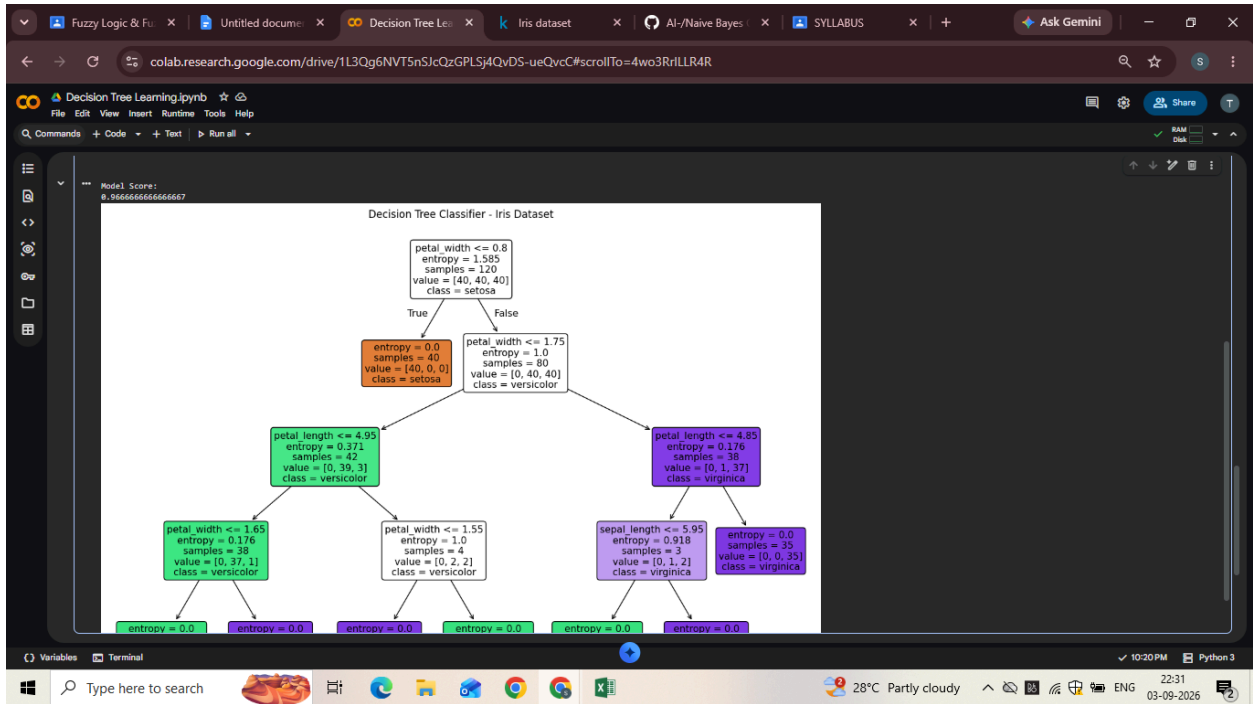
Classification Report:
              precision    recall  f1-score   support

   setosa         1.00        1.00        1.00         10
  versicolor    0.91        1.00        0.95         10
   virginica         1.00        0.90        0.95         10

 accuracy          0.97        0.97        0.97         30
   macro avg       0.97        0.97        0.97         30
  weighted avg     0.97        0.97        0.97         30
```

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